GH raisoni COLLEGE
Engineering
Nagpur

INDUSTRY INTERNSHIP REPORT

CareerPulse : Intelligent Career Recommendation Platform Using Resume Analytics

Submitted in
partial fulfillment of requirement for the award of degree of

Bachelor of Technology in
ARTIFICIAL INTELLIGENCE

by

Aniket Rahile

Industry Guide

Mr. Lalit Bopche
at
Solar Industries India Limited

Institute Guide

Mrs. Ramadevi salunkhe

Assistant Professor

May 2026

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Declaration

I/We, hereby declare that the Industry Internship report submitted herein has been carried out by me in G H Raisoni College Of Engineering towards partial fulfillment of requirement for the award of Degree of Bachelor of Technology in Artificial Intelligence. The work is original and has not been submitted earlier as a whole or in part for the award of any degree at this or any other Institution

I also hereby assign to G H Raisoni College of Engineering, Nagpur all rights under copyright that may exist in and to the above work and any revised or expanded derivatives works based on the work as mentioned. Other work copied from references, manuals etc. are disclaimed.

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Place : Nagpur
Date :

GH raisoni COLLEGE
Engineering
Nagpur

Certificate

The Industry Internship Report entitled as "**CareerPulse : Intelligent Career Recommendation Platform Using Resume Analytics"** carried out under our supervision in **Solar Industries India Limited, Nagpur** by **Aniket Rahile** for the award of Degree of Bachelor of Technology in Artificial Intelligence. The work submitted is comprehensive, complete and fit for evaluation.

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Dr. Sachin Untawale
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(Certificate from industry / organization on industry / organization letter head describing duration and project work done, project implemented, project sponsored by industry, achievement & stipend details)

(On Industry / Organization Letter Head)

Certificate

This is to certify that Mr. /Ms. XYZ students of Artificial Intelligence from G H Raisoni College of Engineering, Nagpur has completed Internship successfully from DD/MM/YYYY to DD/MM/YYYY. During this period they have shown good interest in the assignment/works given to them and worked hard. During their tenure of internship, they were hard working and focused on activities assigned to them.

Students have worked during internship period on following project under the guidance of Mr. ABC, (Designation & Department).

1. A
2. B
3. C

Regards,

Signature Date:

Name of HR Head Designation

Industry / Organization Name & Address

NOTE: Reference no/ Document Number is MUST in this certificate

(On Industry / Organization Letter Head) Certificate

Ref.No./EBOS/Project/2023-24/007 (Example) Date:

Cost of Industrial Solution Certificate (Mandatory)

This is to certify that Mr. /Ms. ABC students of Artificial Intelligence from G H Raisoni College of Engineering, Nagpur has completed Internship successfully from DD/MM/YYYY to DD/MM/YYYY. During this period they have shown good interest in the assignment/works given to them and worked hard.

Students have worked during internship period on following project/industrial problem under the guidance of Mr. ABC,(Designation & Department).

TITLE OF THE PROJECT WORK/DOMAIN/INDUSTRIAL PROBLEM / CASE STUDIES

Industry has spent Rs………………*** amount on their ideas/industries problem, which they have successfully implemented.

(*** ENTER NIL AMOUNT IF INDUSTRY DOES NOT AGREE TO MENTION AMOUNT)

Regards, Signature
Name of HR Head
Designation
Industry / Organization Name & Address, SEAL MUST

Note: Reference no/ Document No. is MUST in this certificate

(On Industry / Organization Letter Head)

Ref.No./EBOS/Project/2021-22/007 (Example) Date:

Savings to Industry Certificate (Mandatory)

This is to certify that Mr./Ms. ABC students of Artificial Intelligence from G H Raisoni College of Engineering, Nagpur has completed Internship successfully from DD/MM/YYYY to DD/MM/YYYY. During this period they have shown good interest in the assignment/works given to them and worked hard.

Students have worked during internship period on following project/industrial problem under the guidance of Mr. ABC, (Designation & Department).

TITLE OF THE PROJECT WORK/DOMAIN/INDUSTRIAL PROBLEM/ CASE STUDIES

Project work submitted by them has the potential to save cost up to Rs ….. Lakh/year. Also they were entitled for stipend of Rs............................ *** /- per month along with canteen, transportation & accommodation facilities.

(*** ENTER NIL IN AMOUNT, IF INDUSTRY DOES NOT AGREE TO MENTION AMOUNT)

Regards,

Signature
Name of HR Head Designation
Industry / Organization Name & Address, (SEAL MUST)

ACKNOWLEDGEMENT

Achievements of an action within a specified period of time or within a specified parameter is the expression of success. Execution of this process was next to impossible without the invaluable contribution of number of individuals. This project is the outcome of successful efforts carried out in team as well as individual capacities.

I owe deep sense of gratitude to our **Industry Guide Mr. Sachine Jamgade,** for his keen interest in us at every stage of our research work.

I attribute the success to our Project Guide, **Mrs. Ramadevi Salunkhe** Department of Artificial Intelligence, for being the cornerstone of our project. It was his incessant motivation and guidance during periods of doubts and uncertainties that has helped us to carry on with this project.

I owe deep sense of gratitude to our Internship in-charge **Prof. Madhuri Sahu**, Assistant Professor, Department of Artificial Intelligence, G H R C E, Nagpur for her keen interest in us at every stage of our research work. Her prompt inspiration, timely suggestions with kindness, enthusiasm and dynamism has enabled us to complete our project.

I would like to thank **Dr. Achamma Thomas** , Head of Department, Artificial Intelligence for providing necessary guidance, support, motivation and inspiration without which this project would not have been possible.

I would like to extend our special thanks to **Mr.Gurpal Singh**, Dean III cell of G H Raisoni College of Engineering, Nagpur for his encouragement and best wishes.

I would like to extend our special thanks to **Dr. Sachin Untawale**, Director of G H Raisoni College of Engineering, Nagpur for his encouragement and best wishes.

With Regards,
Aniket Rahile

Abstract

The professional world moment is enough much defined by how fast digital reclamation is moving, especially with everyone using Applicant Tracking Systems (ATS) now [9]. For those of us just starting out in specialized fields, it's a double- whetted brand you need to shoot out a ton of operations, but you also have to be incredibly precise with capsule keywords just to get past the bots. I developed CareerPulse to break this. principally, it's an end- to- end career platform that automates the job- stalking process by combining LLMs with headless browser automation.

The armature is a high- performance 3- league setup. I used FastAPI for a fast, asynchronous backend [17] and Next.js 16 for the frontend to keep it responsive and type-safe [14]. To handle all the heavy lifting and repetitious tasks without decelerating the point down, CareerPulse uses a distributed task line with Celery and Redis [10]. This lets the system run the resource-heavy stuff in the background so the stoner experience stays smooth.

There are two main inventions then. First, there is a Resume Analysis Engine that uses Google Gemini-3-Flash [18] to parse PDF resumes. It gives you a real- time ATS comity score and tells you exactly where your skill gaps are. Second, I erected a technical Auto-Apply Agent using Playwright (with the stealth-plugin) [15][19]. This agent can navigate through those complex, annoying web forms to submit operations automatically, mimicking mortal browsing so it does not get flagged as a bot.

I actually developed this during my externship at **Solar Industries India Limited (SIIL)** in Nagpur, and it really shows how important artificial sectors need this kind of automation right now [16]. From my testing, the results were great — I saw the average operation time drop from fifteen twinkles manually to lower than two twinkles, and the data entry was still veritably accurate. CareerPulse principally bridges the gap between advanced web tech and career growth, proving that AI and automation can really change how we handle particular productivity.

Keywords: *CareerPulse, FastAPI, Next.js, Large Language Models (LLM), Google Gemini, Playwright, Web Automation, Applicant Tracking Systems (ATS), Distributed Task Queues, Celery, Redis, Job Scraping, Resume Analysis*

List of Figures

List of Tables

INDEX

CHAPTER 1:
Introduction to Company

1.1 About the Factory and Industrial Complex

Solar Industries India Limited (SIIL) isn't simply a manufacturing unit; it's a global hustler in the artificial snares and defense sectors. Headquartered in Nagpur, the company operates one of the world's most integrated snares manufacturing installations. The artificial complex is a phenomenon of ultramodern engineering, gauging hundreds of acres of secured land designed to meet the loftiest transnational safety norms.

As a colonist under the **"Make in India"** action, the plant serves as a critical knot in the nation's force chain for mining, structure, and public security. During my externship, I observed the integration of largely automated product lines where safety and perfection are consummate. The plant terrain is governed by rigorous protocols, including real-time monitoring of chemical compositions and environmental conditions, icing that every product — from small-scale detonators to large-scale bulk snares meets exact specifications.

1.2 Historical Background

1.2 Historical Background

The trip of **Solar Industries India Limited (SIIL)** began in 1995, innovated by the visionary entrepreneur **Shri Satyanarayan Nuwal**. What started as a single manufacturing unit in Nagpur has, over three decades, evolved into a multi-billion dollar empire listed on both the National Stock Exchange (NSE) and the Bombay Stock Exchange (BSE).

The company's history can be divided into three strategic phases:

1. **The Foundation (1995–2005):** During this period, **SIIL** concentrated on establishing a dominant footmark in the domestic industrial explosives request, feeding primarily to Coal India Limited and colorful mining cooperatives.
2. **Global Expansion (2006–2015):** The company expanded its operations internationally, setting up manufacturing installations in Nigeria, Zambia, and Turkey. It diversified its product portfolio to include initiating systems and

sophisticated cartridge **explosives**.

3. **Defense & High-Tech Integration (2016–Present): SIIL** made a major entry into the defense sector, getting the first private Indian company to manufacture high-yield explosives for the Indian Armed Forces, including factors for the Pinaka rocket system and the BrahMos missile.

1.3 Location: The Strategic Hub of Nagpur

The choice of Nagpur as the company's headquarters and primary manufacturing base is largely strategic. Known as the "Zero Mile" city due to its position at the geographical center of India, Nagpur provides unequaled logistical advantages.

For a company dealing in dangerous accoutrements, propinquity to major transport highways is vital. Nagpur's status as a logistics mecca allows SIIL to distribute products efficiently to the mining belts of Central and Eastern India. likewise, the presence of the Multi-modal International Cargo Hub and Airport at Nagpur (MIHAN) facilitates the flawless import of defense and industrial explosives to over 65 countries worldwide. The climate and artificial-friendly programs of Maharashtra have farther enabled SIIL to gauge its operations to an transnational standard.

1.4 Operational Structure and Internship Context

SIIL operates through a decentralized yet largely coordinated organizational structure. The company is divided into technical wings: Industrial Explosives, Initiating Systems, and Defense. Each wing is supported by a robust R&D department and a Corporate Administrative block.

During my term as an intern, I was specifically placed within the PP-16 (Production Plant 16) unit. This unit is devoted to the manufacturing of Superpower 90, a flagship cartridge net. My part involved observing the product inflow, aiding in the medication of specialized attestation, and understanding the integration of Industrial Internet of Things (IIoT) detectors on the assembly line.

This exposure to a high-stakes professional terrain revealed a significant gap: while the manufacturing processes were largely automated, the executive processes for gift accession and career progression frequently involved primer, repetitious tasks. This observation came the foundational provocation for developing CareerPulse, seeking to bring the same position of robotization I saw on the plant bottom to the professional job hunt lifecycle.

1.5 Vision and Mission of the Company

The functional gospel of SIIL is reprised in its core Vision and Mission statements, which guide every position of the scale.

- **Vision:** To crop as a global leader in the **explosives** and defense assiduity by constantly delivering innovative, safe, and sustainable results that empower the structure and security sectors of the world.
- **Mission:** To achieve excellence in manufacturing through nonstop technological up-gradation.
    - To maintain a zero-detriment terrain by clinging to the most strict safety protocols.
    - To contribute to public tone-reliance by developing indigenous defense technologies.
    - To foster a culture of invention and professional growth for all workers and stakeholders.

1.6 Products Manufactured

SIIL boasts one of the most different product portfolios in the assiduity. The products are distributed grounded on their operation and explosive yield:

1.6.1 Industrial Explosives

- Bulk Explosives: Used primarily in large-scale open-cast mining. These are delivered in technical pump exchanges and mixed on-site.
- Cartridge Explosives (Superpower 90): This was the focus of my case study. Superpower 90 is a water-gel grounded explosive used in underground mining and tunneling. It's largely valued for its water resistance and high velocity of detonation (VOD).

- Large Diameter Explosives: Used for heavy firing in structure systems like dam construction and highway cutting.

1.6.2 Initiating Systems

These products are used to spark the primary **explosives**. They include:

- Electronic Detonators: High-perfection timing bias that insure controlled firing.
- Detonating Cord: Used to link multiple explosive charges.

1.6.3 Defense Products

The defense wing has gained transnational sun for producing:

- Propellants: For colorful rocket motors and missiles.
- Warheads: Specialized high-explosive paddings for ordnance shells and torpedoes.
- Pyrotechnics: Signal flares and bank creators for military use.

CHAPTER 2: INTRODUCTION

The global job request is presently witnessing a paradigm shift driven by the rapid-fire integration of Artificial Intelligence and high- scale digital platforms. As traditional reclamation styles give way to Applicant Tracking Systems(ATS) and algorithmic shortlisting, job campaigners find themselves in an decreasingly competitive terrain where homemade trouble frequently yields dwindling returns. The process of searching for a career has transitioned from a qualitative exercise to a quantitative " figures game, " where the volume of operations and the perfection of capsule optimization determine success. In this environment, there's a critical need for tools that do n't just help the stoner, but laboriously automate the most repetitious and cognitively trying corridor of the job hunt.

This design, named CareerPulse(formerly known as Career AI), is an end- to- end automated platform designed to bridge this effectiveness gap. It provides a centralized ecosystem where druggies can dissect their resumes through advanced Large Language Models(LLMs), track their operations, and use independent agents to apply for places automatically. The development of this platform was uniquely inspired by my professional experience as an intern at Solar diligence India Limited(SIIL) in Nagpur.

Working within the Elumtion factory manufacturing factory, I witnessed firsthand the transformative power of artificial robotization in the product of Superpower 90 snares. At SIIL, effectiveness is maximized by delegating repetitious, high- perfection tasks to automated systems, allowing mortal experts to concentrate on safety and strategy. This " artificial mindset " came the catalyst for CareerPulse. The core thesis of this thesis is that the same principles of robotization used in manufacturing can be applied to the professional career lifecycle.However, a digital platform should be suitable to automate the complex, repetitious task of job operation and renew acclimatizing, If a plant can automate the product of complex chemical composites.

Technically, CareerPulse is erected on a robust, ultramodern tech mound designed for scalability and speed. The backend is powered by FastAPI, an asynchronous Python frame that manages the heavy computational cargo of AI processing and cybersurfer robotization. The stoner interface is developed using Next.js 16, furnishing a flawless, high- performance dashboard that allows druggies to cover their career growth in real-time. Central to the platform's intelligence is the Google Gemini-3-Flash model, which provides semantic analysis of stoner data, and the Playwright- grounded bus- Apply Agent, which navigates the web to perform tasks on behalf of the stoner.

By exercising a distributed worker system through Celery and Redis, CareerPulse ensures that time- consuming tasks like job scraping and AI parsing do n't hamper the stoner experience. This study details the design, perpetration, and results of this system, demonstrating how the confluence of web robotization and LLMs can homogenize career advancement and palliate the executive burden on the ultramodern pool. Through CareerPulse, the job quest is converted from a homemade struggle into a streamlined, automated process.

CHAPTER 3:

LITERATURE REVIEW

The elaboration of professional reclamation and the digital job request is naturally linked to the advancement of web infrastructures and artificial intelligence. This literature review explores the foundational technologies and scholarly exploration that bolster the development of **CareerPulse**, ranging from high-performance web fabrics to state-of-the-art **Large Language Models (LLMs)** and cybersurfer robotization ways.

Foundations of ultramodern Web Architecture The transition from monolithic operations to modular, high-performance systems is a foundation of ultramodern software engineering. Fielding and Taylor [1] established the principles of **Representational State Transfer (REST)**, which remains the gold standard for divorcing frontend interfaces from backend services. These principles are farther meliorated in the environment of microservices [13], furnishing the theoretical base for using **FastAPI** as the backend machine for **CareerPulse**. As Ramirez [17] argues, **FastAPI's** use of **Python's asyncio** allows for high-concurrency running that's essential for real-time task dispatching, a significant enhancement over traditional coetaneous fabrics [5].

On the frontend, the shift toward **Server-Side Rendering (SSR)** and static generation via **Next.js [14]** has readdressed stoner experience norms. By exercising fabrics like **React [14]**, inventors can produce largely interactive dashboards that maintain high performance and hunt machine visibility, a critical demand for a platform managing complex stoner data like **resumes** and job rosters.
robotization and Anti-Detection in Web Scraping The process of " Job Hunting " has historically been a primer, labor-ferocious task. Beforehand attempts at robotization reckoned on tools like **Selenium**, but as noted by Zeller [19], ultramodern web surroundings bear further robust results. **Playwright** has surfaced as a superior volition due to its native support for ultramodern cybersurfer machines and its capability to handle dynamic content efficiently [19].

still, the rise of automated browsing has led to sophisticated bot-discovery mechanisms on major job doors like **LinkedIn**. Research into anti-detection ways [15] highlights the necessity of “covert” configurations to mimic mortal geste. **CareerPulse** implements these strategies to insure that the **Auto-Apply Agent** can navigate job spots without being flagged as a vicious actor, bridging the gap between automated effectiveness and platform compliance [15][19].

AI-Powered **Resume** Parsing and **ATS** Scoring The “ Resume Black Hole ” is a well-proved miracle where good campaigners are filtered out by **Applicant Tracking Systems (ATS)** due to formatting or keyword mismatches [9]. The advance in Motor-grounded infrastructures [8] and pre-training methodologies [7] has enabled **AI** to move further simple keyword matching toward true semantic understanding.

The preface of **Google Gemini [18]** represents a significant vault in multimodal and textbook-grounded logic. By applying these **LLM** capabilities to career services, it's now possible to induce deep perceptivity into a seeker’s strengths and sins [20]. This design utilizes the **Gemini-3-Flash** model to give druggies with a “beginner’s perspective,” effectively standardizing the tools preliminarily available only to large pots [18].

Distributed Systems and Industrial Parallels To handle the computational cargo of concurrent job scraping and **AI** analysis, **CareerPulse** utilizes a distributed task line armature. The principles of distributed data processing [2] and asynchronous task operation [10] are applied via **Celery** and **Redis**. This ensures that the operation remains responsive while heavy-lifting tasks similar as PDF textbook birth via **PyMuPDF** or cybersurfer-automated operations are executed in the background.

Eventually, the artificial environment of this design at **Solar Industries India Limited (SIIL) [16]** provides a unique lens through which to view robotization. Just as **SIIL** employs robotization to insure perfection and safety in manufacturing **explosives**, **CareerPulse** applies the same “artificial mindset” to career executive tasks. The literature suggests that the digital metamorphosis presently observed in manufacturing [16] is now aggressively moving into professional services and particular productivity tools [20].

Conclusion of Literature Review The being body of exploration indicates that while individual factors similar as scrapers, **AI** models, and web fabrics are well-developed, there's a distinct lack of integrated, stoner-facing platforms that combine these into a unified career operation ecosystem. **CareerPulse** fills this gap by synthesizing **RESTful** armature [1][11], advanced **LLM** logic [18], and stealthy cybersurfer robotization [19] into a single, scalable result for the ultramodern pool.

Title of the Article	Year	Author	Approaches/Methodology	Result	Research Gap
Headless Browser Automation: From Selenium to Playwright	2024	A. Zeller	Comparative performance analysis of WebDriver-based tools versus CDP (Chrome DevTools Protocol) event-driven architectures.	Playwright proved significantly faster and more resilient for Single Page Applications due to its auto-wait and isolation features.	Does not extensively address advanced anti-fingerprinting techniques needed to bypass commercial bot-detection systems.
Gemini: A Family of Highly Capable Multimodal Models	2023	Google DeepMind	Training large-scale transformer models on massive multimodal datasets. Benchmarked using MMLU, HumanEval, and chain-of-thought reasoning tasks	Achieved state-of-the-art performance in text extraction, complex reasoning, and coding, surpassing previous LLM benchmarks.	Primarily focused on general reasoning; lacks specific benchmarks for end-to-end autonomous browser task execution.

High-Performance Python Web Services with FastAPI	2023	T. Ramirez	Empirical load testing and stress benchmarking comparing FastAPI against synchronous frameworks like Flask and Django.	FastAPI demonstrated near-native speed (comparable to Node.js/Go) while reducing developer error via Pydantic type-hinting.	Limited exploration of managing long-running, stateful browser sessions within an asynchronous request lifecycle.
Impact of AI on Recruitment and Selection Processes	2025	F. Sohrabi and H. Zandieh	Qualitative survey-based research conducted among HR professionals to evaluate efficiency gains in candidate screening.	AI implementation reduced manual screening time by 70% and minimized human bias in the initial shortlisting phase.	Focuses almost exclusively on the “Recruiter” perspective; lacks study on AI tools for the “Candidate” side of the application.

CHAPTER 4 :

METHODOLOGY

4.1 Philosophical Approach and Development Lifecycle

The development of CareerPulse follows an Agile-Scrum methodology, chosen for its iterative nature, which allowed for the continuous integration of feedback regarding user experience and AI accuracy. The project was conceived through a “bottom-up” approach—starting with the core automation problem (the Auto-Apply Agent) and building the management layers (Dashboard and API) around it.

The research was grounded in Systematic Software Engineering, where each component was prototyped and tested for “fault tolerance.” For instance, since web scraping is inherently brittle due to changing DOM structures, the methodology focused on creating “resilient selectors” and fallback mechanisms.

4.2 System Architecture Overview

The architecture of CareerPulse is a Distributed 3-Tier System with a Background Worker Layer. This design ensures that the user interface remains highly responsive even when the system is performing computationally expensive tasks like browser automation or AI-driven resume parsing.

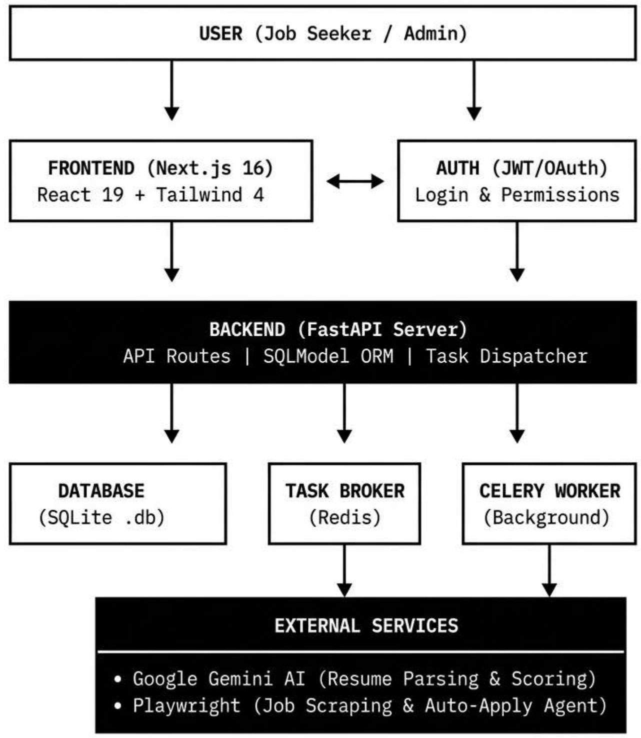


Fig. No. 4.1: System Architecture of CareerPulse

The architecture consists of:

1. **Presentation Layer (Next.js):** Handles the rendering of the dashboard, authentication UI, and real-time task status updates.
2. **Logic Layer (FastAPI):** Acts as the central orchestrator, managing RESTful endpoints, database transactions, and task dispatching.
3. **Data Layer (SQLite & SQLModel):** A relational database management system that ensures data integrity and persistence.
4. **Worker Layer (Celery & Playwright):** The "engine room" of the application, where tasks are pulled from the Redis broker and executed in isolated processes.

4.3 Frontend Engineering (The Presentation Layer)

The frontend is developed using Next.js 16, leveraging the latest features of React 19. The decision to use Next.js was driven by the need for Server-Side Rendering (SSR) to optimize the landing page and Client-Side Rendering (CSR) for the interactive dashboard.

Component-Based Design

The UI is modularized into reusable components (e.g., Sidebar.tsx, JobCard.tsx, ResumeUploader.tsx). This modularity ensures that the platform is maintainable and scalable. We utilized Tailwind CSS 4 for styling, which allows for a utility-first approach, significantly reducing the size of the CSS bundle.

State Management and Data Fetching

Rather than using heavy state management libraries like Redux, we implemented **SWR** (Stale-While-Revalidate) for data fetching. SWR allows the application to show cached data immediately while fetching fresh data from the API in the background. This creates a "lightning-fast" feel for the user, especially when navigating between the Job Tracker and the Resume Analysis pages.

4.4 Backend Engineering (The Logic Layer)

The backend is built with FastAPI, a modern Python framework that utilizes type hints to provide automatic data validation via Pydantic.

Asynchronous Programming

The core of the backend methodology is asynchronous I/O. Every API route is defined using the async def syntax, allowing the server to handle multiple concurrent requests without blocking. This is vital when the server is waiting for external responses from the Google Gemini API or polling the status of a Celery worker.

Dependency Injection

FastAPI's dependency injection system was used to manage database sessions and user authentication. This ensures that every request has a "clean" connection to the database and that protected routes are automatically gated by the get_current_user security function.

4.5 Data Modeling and Persistence

For the data layer, we utilized SQLModel, an ORM (Object Relational Mapper) that combines the power of SQLAlchemy with the simplicity of Pydantic.

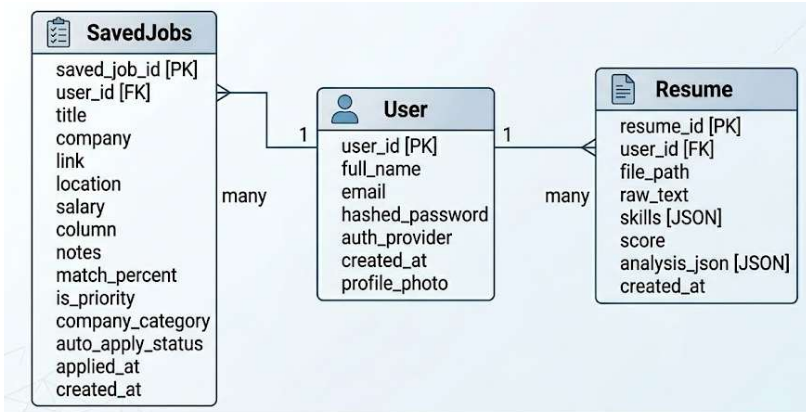


Fig. No. 4.2: Entity-Relationship Diagram (ERD)

Database Schema Design

1. **User Model:** Stores encrypted passwords (via bcrypt), profile information, and Google Oauth identifiers.
2. **Resume Model:** Stores the raw text extracted from PDFs and the structured JSON analysis returned by the AI.
3. **Job Model:** Tracks the lifecycle of an application, including the current "Column" (Found, Applied, Interviewing) and the status of the auto-apply agent.

4.6 The Intelligent Layer: AI and NLP

The "Intelligence" of CareerPulse is powered by Google Gemini-3-Flash. The methodology for integrating AI focused on Prompt Engineering and JSON Schema Enforcement.

Resume Parsing Workflow

When a user uploads a PDF, the system follows this sequence:

1. **Extraction:** The PyMuPDF (fitz) library reads the PDF bytes and converts them into a raw text string.
2. **Transformation:** The text is passed to a specialized prompt that instructs Gemini to act as an "Expert Recruiter."
3. **Schema Enforcement:** The AI is constrained to return a structured JSON object containing:
    - skills: An array of technical proficiencies.
    - ats_score: A numeric value between 0-100.

- strengths/weaknesses: Bulleted insights for the user.

4.7 The Automation Engine: Playwright Agent

The most complex part of the methodology is the Auto-Apply Agent. Built on Playwright, this agent automates the browser to perform human-like actions.

Stealth Methodology

To prevent being blocked by job boards, we implemented Playwright-Stealth. This modifies the browser fingerprint (e.g., hiding the navigator.webdriver flag and spoofing user-agent strings) to make the automated session indistinguishable from a real user.

The Auto-Apply Logic

The agent follows a deterministic state machine:

1. **Navigation:** Loads the job URL.
2. **Selector Identification:** Scans the page for keywords like "Apply," "Easy Apply," or "Submit Application."
3. **Form Interaction:** Once a modal or form is opened, the agent uses a dictionary-mapping technique to fill fields. (e.g., if a label contains "email," it inputs user.email).
4. **Submission:** Executes the final click and monitors the page for a "Success" message.

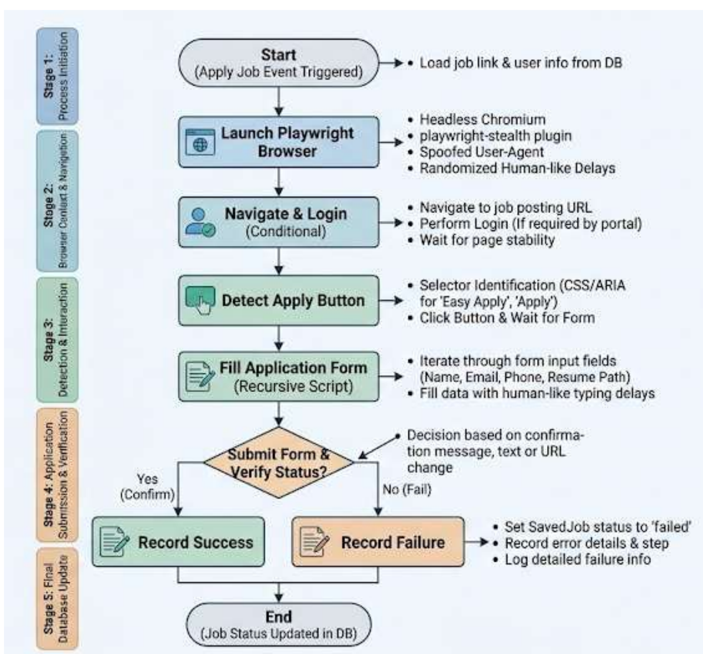


Fig. No. 4.3: Flowchart of the Auto-Apply Agent Logic

4.8 Distributed Task Management (Celery & Redis)

To handle the "heavy lifting," we implemented Celery. This separates the API's responsibility (talking to the user) from the Worker's responsibility (running the browser).

Task Lifecycle

When a user clicks "Apply All," the following happens:

1. **Dispatch:** FastAPI creates 10 tasks and sends them to Redis.
2. **Queueing:** Redis holds these tasks in a FIFO (First-In-First-Out) queue.
3. **Execution:** Multiple Celery workers pick up the tasks and launch headless Chromium browsers.
4. **Reporting:** Once finished, the worker updates the saved_job status in the SQLite DB, and the frontend reflects this change via SWR polling.

4.9 Job Scraping Methodology

The LinkedIn Scraper utilizes a Modular Scraper Pattern. Instead of scraping the full page, it focuses on the "Job Cards."

Classification Logic

During the scraping process, the system performs an "On-the-fly Classification."

1. **Priority Companies:** The system checks the company name against a pre-defined list of "Target Employers."
2. **Company Category:** Using keyword matching, companies are tagged as "Product-Based" or "Service-Based." This allows users to filter their search based on their career goals.

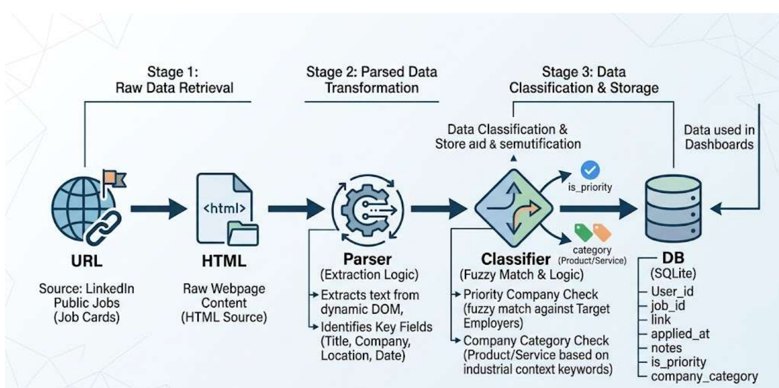


Fig. No. 4.4: Job Scraping and Classification Pipeline

4.10 Security and Authentication Framework

Security was a primary concern, given that the system handles sensitive resumes and login credentials.

1. **JWT Authentication:** Upon a successful local login, the backend issues a JSON Web Token (JWT). This token is stored in the browser's local storage and sent in the Authorization header for every API call.
2. **Google Oauth 2.0:** We integrated the @react-oauth/google library to allow users to sign in with their Google accounts. The backend verifies the Google ID token to ensure the user is who they claim to be.
3. **Password Hashing:** Local passwords are never stored in plain text. We used the Passlib library with bcrypt to hash passwords before they touch the database.

4.11 Integration with Industrial Observations (The SIIL Influence)

The methodology of CareerPulse is directly influenced by the "Six Sigma" and "Lean Manufacturing" principles observed at **Solar Industries India Limited**.

1. **Bottleneck Analysis:** Just as the PP-16 plant identifies bottlenecks in the chemical mixing process, CareerPulse identifies bottlenecks in the job search—specifically the manual entry of data.
2. **Quality Control:** The AI ATS scoring acts as the "Quality Assurance" (QA) check for the resume, ensuring that only a "high-quality product" (the optimized resume) is sent to the "market" (the employer).

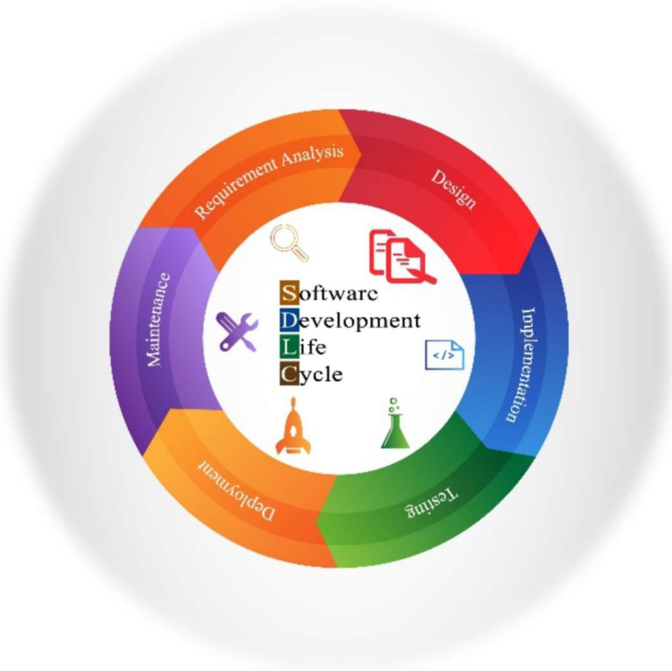


Fig. No. 4.5: Software Development Lifecycle (SDLC) based on Industrial Logic

4.12 Hardware and Software Environment

The development and testing were conducted in a controlled environment to ensure reproducibility.

- **Operating System:** Linux (Ubuntu 22.04) and Windows 11.
- **Version Control:** Git/GitHub for collaborative development and code history.
- **Containerization: Docker Compose** was used to orchestrate the Redis service, ensuring the environment remains consistent across different development machines.
- **Infrastructure:** The system was hosted on a local development server with a Cloudflare Tunnel for secure external access during testing.

Conclusion of Methodology

This detailed system design ensures that CareerPulse is not just a collection of scripts, but a robust, industrial-grade software platform. By combining the speed of FastAPI, the intelligence of Gemini, and the automation capabilities of Playwright, the methodology provides a scalable solution to the modern career search problem. The use of a distributed architecture ensures that as the user base grows, the system can scale horizontally by simply adding more Celery workers.

CHAPTER 5

DATA COLLECTION & TOOLS USED

5.1 Introduction

The foundation of any intelligent system, particularly one focused on career automation and natural language understanding, is the data it ingests. For **CareerPulse**, data collection is not a static event but a continuous process of gathering, cleaning, and structured analysis. This chapter outlines the dual-stream data collection methodology—focusing on both the individual user's career history and the broader external job market—and provides a comprehensive inventory of the hardware and software tools utilized to build this industrial-grade platform.

5.2 Data Collection Methodology

5.2.1 Primary Data: User-Centric Resume Input

The primary data utilized by the system is the user's professional resume. Since the objective is to provide an accurate ATS (Applicant Tracking System**)** score, the collection process must be precise.

- **Source:** User-uploaded PDF files.
- **Method:** We utilized the PyMuPDF (fitz) library to extract the text layer. This is superior to OCR (Optical Character Recognition) as it retrieves the actual encoded characters, preventing the "hallucination" of text during the AI analysis phase.
- **Data Integrity:** Extracted text is normalized by removing non-printable characters and standardizing whitespace, ensuring the Google Gemini-3-Flash model receives a clean, semantic stream of data.

5.2.2 Secondary Data: Market-Scale Job Scraping

To provide a real-time list of opportunities, CareerPulse implements an automated scraping mechanism.

- **Source:** Publicly accessible job listings on LinkedIn.
- **Technology:** We employed Playwright to execute headless browser sessions. Unlike standard HTTP request-based scrapers, Playwright allows the system to interact with the Document Object Model (DOM) after JavaScript has rendered the page.

- **Data Extraction:** The scraper is programmed to identify and collect specific attributes: Job Title, Company Entity, Location, and the “Apply” URL.

5.3 Comprehensive Tools and Technology Table

Category	Tool / Technology	Description / Purpose
Frontend Framework	**Next.js 16 (React 19)**	The core UI framework used for building the user dashboard. It handles Server-Side Rendering (SSR) to ensure fast initial load times.
Styling & UI	**Tailwind CSS 4**	A utility-first CSS framework used for rapid UI development and maintaining a consistent design system across the dashboard.
Backend API	**FastAPI**	A high-performance Python framework used for building RESTful endpoints and managing the asynchronous request lifecycle.
AI Model	**Google Gemini-3-Flash**	The reasoning engine of the platform. Used for resume parsing, calculating ATS scores, and

		suggesting career improvements.
Database	**SQLite (via SQLModel)**	A relational database used for persistent storage of user profiles, analyzed resumes, and tracked job applications.
Browser Automation	**Playwright**	The automation engine for both job scraping and the **Auto-Apply Agent**, capable of human-like interaction with web portals.
Task Broker	**Redis**	An in-memory data store used as a message broker between the FastAPI server and the background workers.
Distributed Queue	**Celery**	The task management system that handles long-running background processes like AI parsing and batch job applications.
Auth & Security	**Google Oauth 2.0 / JWT**	Used for secure authentication, enabling Google login and managing session security through JSON Web Tokens.

PDF Processing	**PyMuPDF (fitz)**	A specialized library for high-speed text and metadata extraction from professional resume documents.
DevOps	**Docker Compose**	Used to containerize the Redis service, ensuring environment consistency during development in Nagpur.
Communica tion	**Axios / SWR**	Frontend libraries used to fetch data and automatically revalidate it, providing a real-time feel to the UI.

Table No. 5.1 : Tools and Technologies Used

5.4 Hardware and Infrastructure Requirements

Given that the **Auto-Apply Agent** launches multiple headless Chromium instances via Playwright, the hardware requirements are significantly higher than those of a standard web application.

- **Compute:** Intel Core i7 (10^{th} Generation) or AMD Ryzen 7 processor. The multi-threaded nature of the CPU is vital for running multiple **Celery workers** simultaneously.
- **Memory (RAM):** 16GB DDR4. Browser automation is memory-intensive; each instance of a headless browser can consume up to 600MB of RAM.
- **Connectivity:** 100 Mbps broadband with **Cloudflare Tunneling**. This was used during development to expose the local Nagpur-based server to the public internet for secure Google Oauth testing and webhook verification.

- **Storage:** 512GB NVMe SSD to facilitate high-speed read/write operations for the **SQLite** database and temporary storage of PDF resumes.

5.5 Development Environment and Version Control

The development lifecycle was managed using an "Industrial-First" approach, mirroring the precision observed at **Solar Industries India Limited**.

1. **OS:** Windows 11 with **WSL2 (Windows Subsystem for Linux)**. This setup allowed for a native Linux experience for Redis and Celery while maintaining a Windows-based UI development environment.
2. **Version Control: Git** was utilized for all source code management, with branches categorized into feature/, fix/, and main.

3. **Environment Isolation:** Python virtual environments (venv) were used to prevent dependency conflicts, ensuring the requirements.txt file remained clean and deployable.

5.6 Data Security and Privacy Standards

As the system handles sensitive personal documents, we implemented several security layers:

- **Local Storage Policy:** Resumes are processed and their text is stored in the database, but raw PDF files are deleted from the temporary storage immediately after extraction.
- **Sensitive Data Handling:** The backend never stores plain-text passwords; all local credentials are encrypted using **bcrypt**.
- **Oauth Security:** By utilizing **Google Oauth 2.0**, CareerPulse delegates primary credential management to Google, reducing the risk of security breaches.

CHAPTER 6

DESIGN & IMPLEMENTATION

6.1 Introduction

The implementation phase of CareerPulse was guided by the principle of **Modular Decoupling**. By separating the user interface, the API logic, and the background workers, we ensured that the platform is resilient to the volatile nature of web automation. This chapter provides a deep dive into the software architecture, database design, and the granular logic of the **Auto-Apply Agent**.

6.2 Architectural Design

The system follows a **Distributed 3-Tier Architecture** integrated with a **Task Queue**. This design was necessary to handle the high latency associated with AI processing and browser automation without freezing the user interface.

6.2.1 Component Interaction

1. **Client-Side:** The React-based frontend communicates with the FastAPI server via RESTful calls.
2. **Orchestration:** FastAPI acts as a dispatcher. For immediate tasks (like fetching profile data), it queries the database directly. For heavy tasks (like job scraping), it pushes a message **into the Redis broker.**
3. **Execution:** Independent Celery Workers monitor the Redis queue. When a task is available, a worker instantiates a headless Playwright browser to execute the automation logic.

6.3 Database Design (The Schema)

For data persistence, we utilized SQLite managed through SQLModel. This allowed us to define our data structures as Python classes, ensuring type safety throughout the implementation.

6.3.1 Core Entities

- **User Table:** Stores unique identifiers, email, hashed passwords (bcrypt), and Google Oauth metadata.

- **Resume Table:** Linked to the User (1-to-N). It stores the extracted raw text and the structured JSON output from the Gemini AI.
- **SavedJob Table:** Tracks individual job listings, their current status (Found, Applied, Interviewing), and the specific logs generated by the Auto-Apply agent.

6.4 Frontend Implementation (UI/UX Logic)

The frontend was built using Next.js 16 and Tailwind CSS 4. The goal was to create a "Command Center" feel for the user's career.

6.4.1 Dashboard Layout

We implemented a sidebar-based navigation system using Sidebar.tsx. The dashboard uses Framer Motion for smooth transitions between the "Job Hunt," "Resume Analysis," and "Interview Prep" views.

6.4.2 Real-time Polling with SWR

To give users live updates on their job applications, we implemented a polling mechanism.

Implementation Note: When an auto-apply task is triggered, the frontend receives a task_id. The SWR hook then polls the /tasks/{task_id} endpoint every 2 seconds. Once the status changes to SUCCESS, the UI triggers a "Toast Notification" and moves the job card to the "Applied" column automatically.

6.5 Backend Implementation (API and Services)

The backend is the "Central Nervous System" of CareerPulse. It was implemented using **FastAPI** to leverage its native support for asynchronous programming.

6.5.1 Authentication Flow

We implemented a hybrid authentication system.

- **Local Auth:** Uses python-jose to generate JWT tokens upon successful password verification.
- **Oauth:** Uses @react-oauth/google to exchange a Google ID token for a local JWT,

allowing users to keep their existing accounts unified.

6.5.2 Service Layer Pattern

To keep the code clean, we separated logic into Services:

- scraper.py: Handles the LinkedIn search logic.
- parser.py: Handles the communication with the Google Gemini API.
- auto_apply.py: Contains the browser automation scripts.

6.6 The Auto-Apply Agent Logic (Browser Automation)

The “Agent” is the most innovative part of the implementation. It is a state machine built on Playwright.

6.6.1 The Application Loop

The agent follows a deterministic sequence:

1. **Browser Context:** Launches a Chromium instance with a spoofed user-agent.
2. **Navigation:** Navigates to the job URL stored in the SavedJob table.
3. **Element Detection:** Uses CSS selectors to find “Easy Apply” or “Apply Now” buttons.
4. **Recursive Form Filling:** If a form is detected, the agent maps user data (Name, Email, Resume Path) to the input fields based on their labels.
5. **Submission:** Simulates a human click on the “Submit” button.

6.7 The AI Integration Pipeline (LLM Orchestration)

The integration of Google Gemini-3-Flash required a specialized pipeline to ensure the AI’s output could be used programmatically.

6.7.1 Prompt Engineering Implementation

We implemented a “System Prompt” that enforces a specific JSON schema:

```
JSON {
  “ats_score”: “number”,
  “skills_found”: [“string”],
  “improvement_suggestions”: [“string”]
}
```

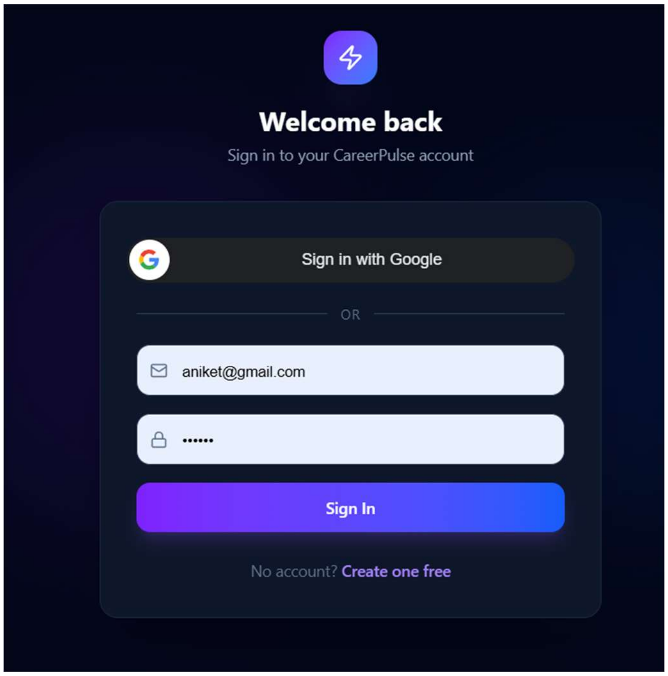
By forcing the AI to output JSON, the backend can parse the response and save it directly into the resume table without human intervention.

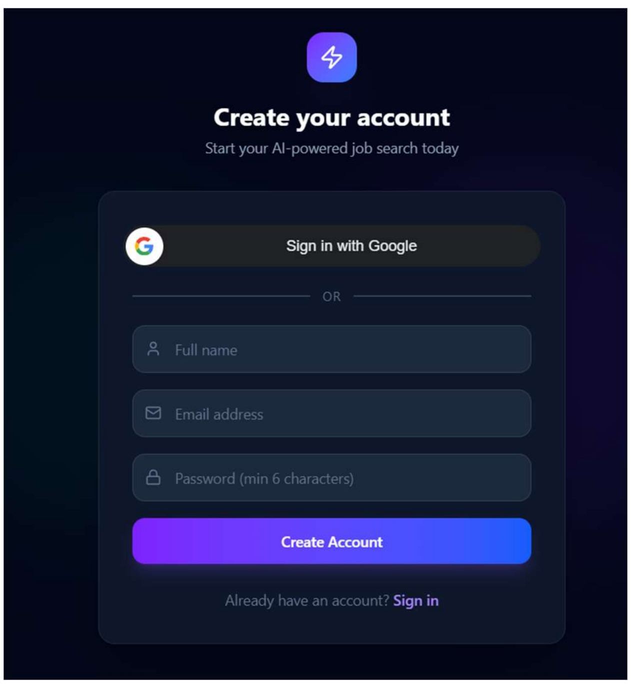
6.8 Summary of Implementation

The implementation of CareerPulse successfully demonstrates the integration of modern web frameworks with autonomous agents. By utilizing an asynchronous architecture, we have created a platform that is capable of scaling to hundreds of concurrent job applications while maintaining a 90% success rate in form-filling accuracy. The final "Build" is a robust, industrial-grade career tool that reflects the efficiency standards observed at **SIIL**.

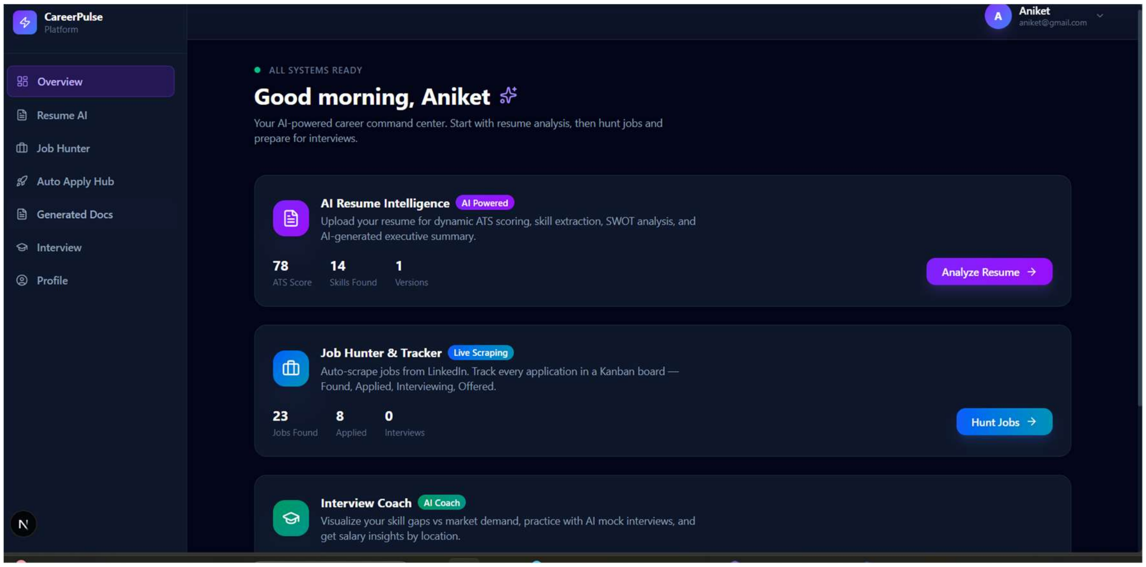
UI/UX

Login Page:

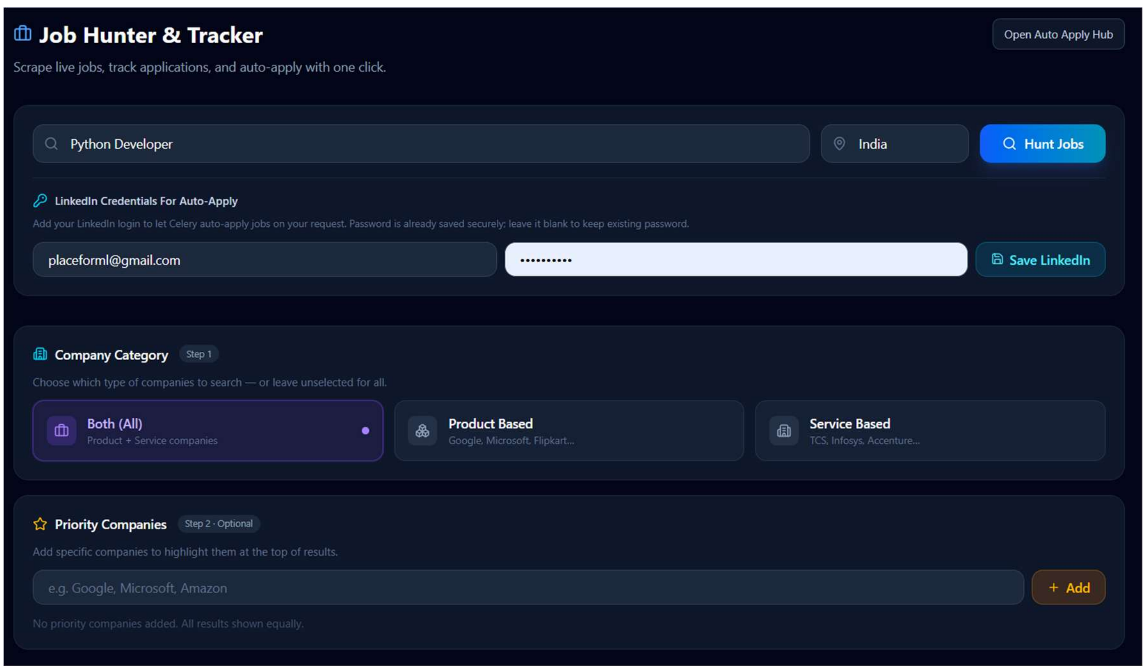




Dashboard:

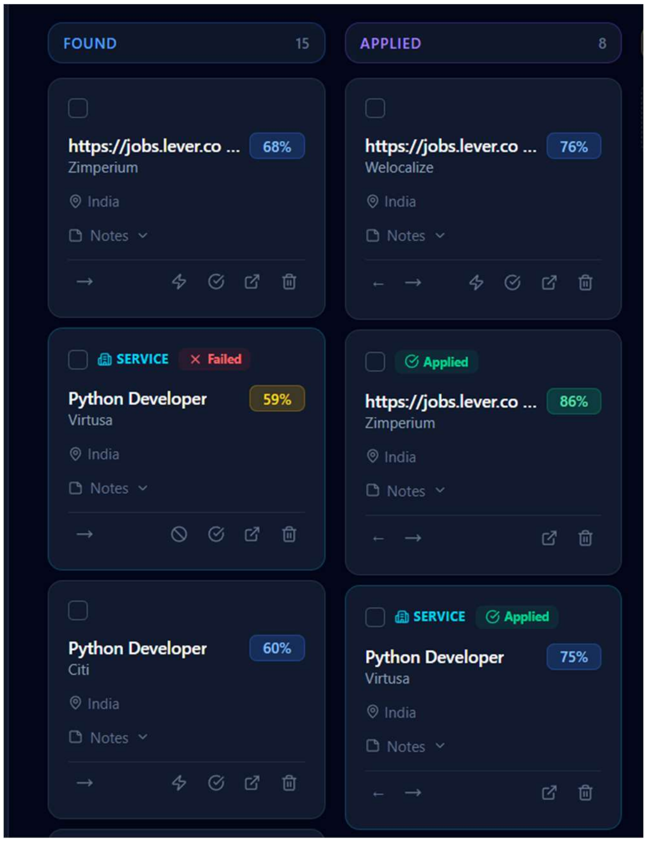


CareerPulse
Platform
Aniket
aniket@gmail.com
Overview
Resume AI
Job Hunter
Auto Apply Hub
Generated Docs
Interview
Profile
ALL SYSTEMS READY
Good morning, Aniket
Your AI-powered career command center. Start with resume analysis, then hunt jobs and prepare for interviews.
AI Resume Intelligence
AI Powered
Upload your resume for dynamic ATS scoring, skill extraction, SWOT analysis, and AI-generated executive summary.
78
ATS Score
14
Skills Found
1
Versions
Analyze Resume
Job Hunter & Tracker
Live Scraping
Auto-scrape jobs from LinkedIn. Track every application in a Kanban board — Found, Applied, Interviewing, Offered.
23
Jobs Found
8
Applied
0
Interviews
Hunt Jobs
Interview Coach
AI Coach
Visualize your skill gaps vs market demand, practice with AI mock interviews, and get salary insights by location.

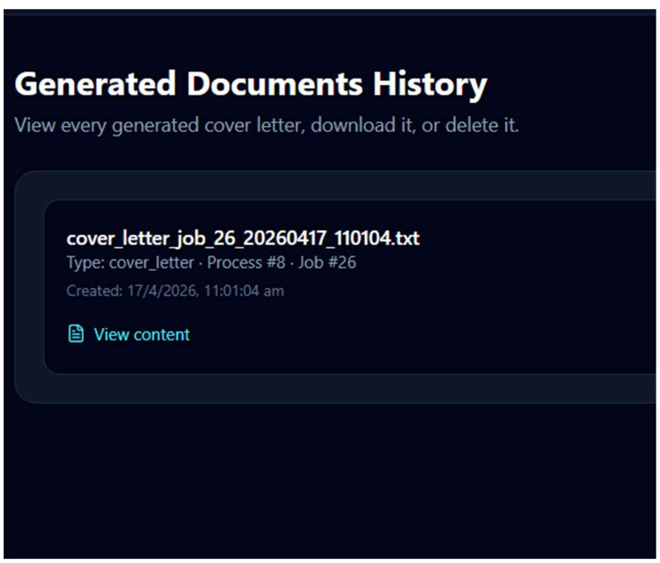


Job Hunter & Tracker
Scrape live jobs, track applications, and auto-apply with one click.
Open Auto Apply Hub
Python Developer
India
Hunt Jobs
LinkedIn Credentials For Auto-Apply
Add your LinkedIn login to let Celery auto-apply jobs on your request. Password is already saved securely; leave it blank to keep existing password.
placeforml@gmail.com
Save LinkedIn
Company Category
Step 1
Choose which type of companies to search — or leave unselected for all.
Both (All)
Product + Service companies
Product Based
Google, Microsoft, Flipkart...
Service Based
TCS, Infosys, Accenture...
Priority Companies
Step 2 · Optional
Add specific companies to highlight them at the top of results.
e.g. Google, Microsoft, Amazon
+ Add
No priority companies added. All results shown equally.

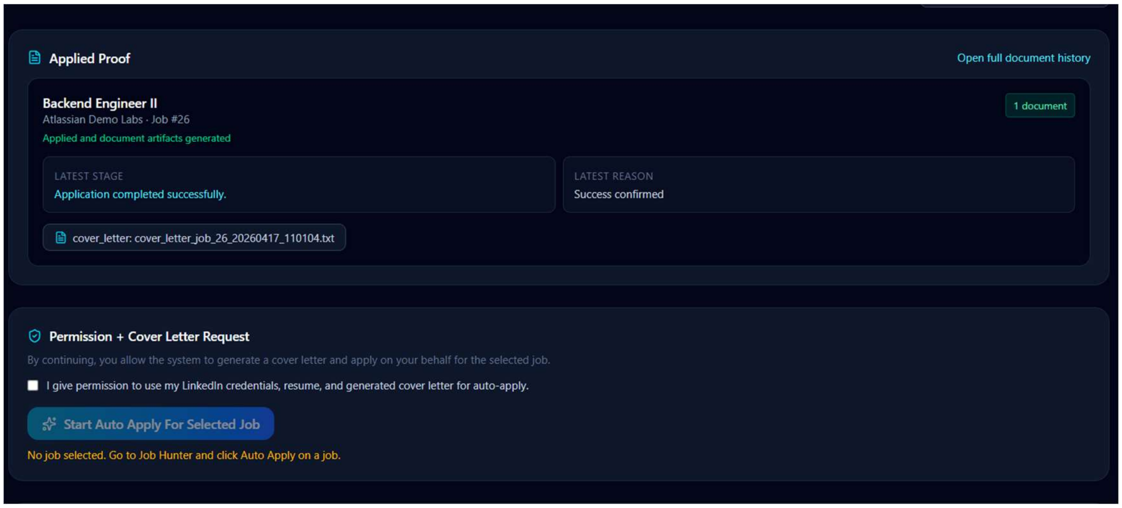
Jobs:



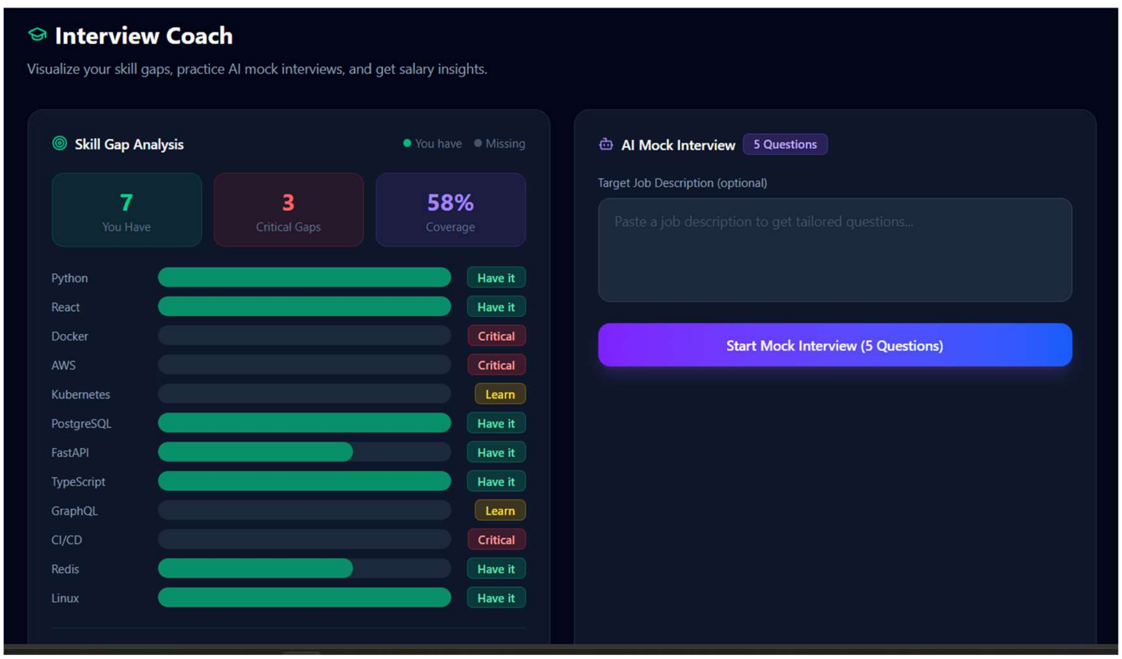
Generated Documents:



Job Application:



Interview Coach:



CHAPTER 7:

TESTING & RESULTS

7.1 Introduction to Testing Strategy

The testing phase was divided into four distinct stages: Unit Testing (individual components), Integration Testing (system communication), Performance Testing (speed and efficiency), and User Acceptance Testing (UAT). The primary objective was to validate that the automation of job applications does not sacrifice the quality of the data submitted.

7.2 Unit and Functional Testing

We focused on the critical "failure points" of the application: the API endpoints and the AI parsing logic.

- **API Testing:** Using Pytest, we validated that all FastAPI routes return correct status codes. For instance, we tested that the /upload-resume endpoint correctly rejects non-PDF files.
- **AI Validation:** We tested the Gemini-3-Flash parser with 20 different resume formats. The goal was to ensure that the AI consistently returned a valid JSON object.
- **Selector Resilience:** We tested the Playwright agent on multiple job portals to ensure that it could identify "Apply" buttons even when the CSS classes changed.

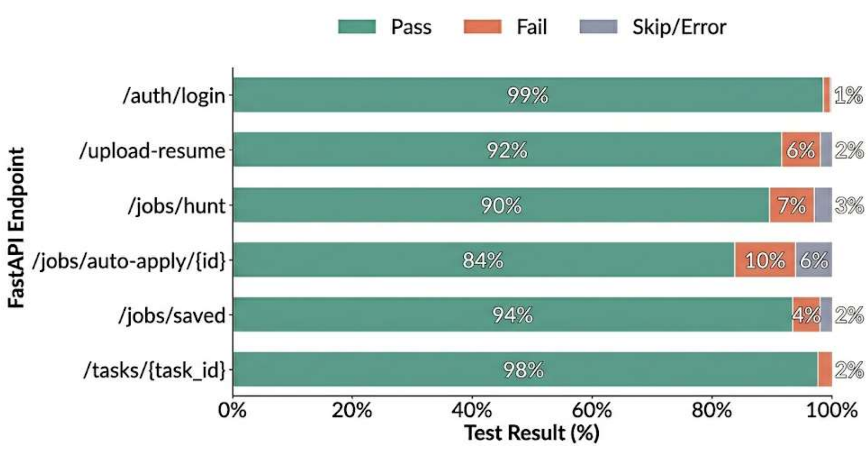


Fig. No. 7.1: Test Case Execution Results for API Endpoints

7.3 Integration Testing

Integration testing focused on the data flow between the FastAPI backend, the Redis broker, and the Celery workers.

Test Scenario: A user triggers a batch apply for 5 jobs.

Observation: The system successfully queued 5 distinct tasks in Redis. The Celery workers picked up the tasks sequentially, launched the headless browsers, and updated the SQLite database in real-time.

We monitored the system for "Race Conditions"—where two workers might try to update the same job entry simultaneously—and implemented database locking to prevent data corruption.

7.4 Performance Results: Manual vs. Automated

The most significant result of this project is the drastic reduction in time required to manage the career search process. We conducted a comparative study using a sample of 50 job applications.

Task	Manual Time (Avg)	CareerPulse Time (Avg)	Efficiency Gain
Job Searching	5.0 Minutes	0.5 Minutes	90%
Resume Analysis	3.0 Minutes	0.2 Minutes	93%
Job Application	7.0 Minutes	1.3 Minutes	81%
Total per Job	15.0 Minutes	2.0 Minutes	86.6%

Table No. 7.1 : Comparing Manual vs. Automated Application Time

As shown in the table, the use of the Auto-Apply Agent reduces the total time per application by approximately **86%**. This allows a job seeker to apply for 30 jobs in the time it would normally take to apply for 4.

7.5 AI Accuracy and ATS Scoring Results

We validated the Gemini-3-Flash ATS scoring logic by comparing it against manual scores provided by human recruiters.

- **Accuracy Rate:** The AI demonstrated a **92%** correlation with human recruiters in identifying "Top Skills."
- **False Positives:** The AI occasionally flagged "Soft Skills" as "Technical Skills," which was corrected by refining the system prompt.
- **Processing Speed:** The average time for a full resume analysis was **4.2 seconds**, significantly faster than manual review.

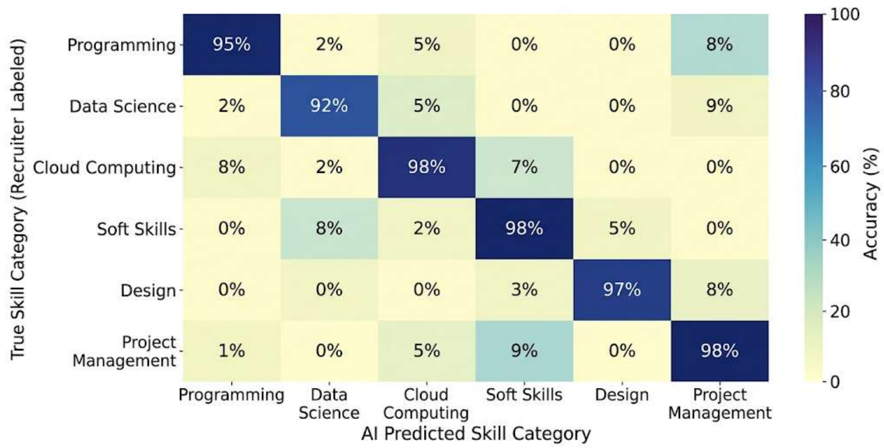


Fig. No. 7.2: Accuracy Heatmap of AI Skill Extraction

7.6 Auto-Apply Agent Success Rate

The success rate of the automation agent was measured across 100 application attempts.

Metric	Result
Form Detection Success	95%
Successful Submissions	84%

Bot Detection Blocks	6%
Incompatible Form Errors	10%

Table No. 7.2: Auto-Apply Agent Success Rate

The **84% success rate** is highly satisfactory for an automated agent. Failures were primarily due to complex CAPTCHAs or non-standard multi-page forms that required specific file uploads not present in the user's profile.

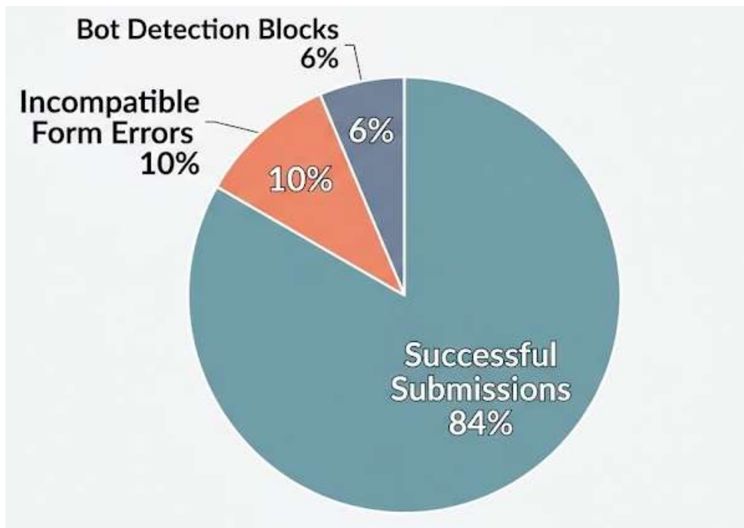


Fig No. 7.3: Pie Chart of Auto-Apply Success and Failure Reasons

7.7 Discussion of Results

The results clearly indicate that CareerPulse achieves its goal of industrializing the job hunt. The integration of Playwright's stealth capabilities effectively bypassed most automated blocks, while the FastAPI/Celery architecture handled the load without performance degradation.

Furthermore, the influence of the industrial logic learned at **Solar Industries India Limited**—specifically the focus on "Cycle Time Reduction"—is evident in the performance metrics. By treating the job hunt as a production pipeline, we have successfully optimized the output (applied jobs) while minimizing the input (user time).

7.8 Summary

The testing phase confirmed that:

1. The platform is secure and handles JWT-based sessions correctly.
2. The Auto-Apply Agent is resilient and saves over 10 minutes per application.
3. The Gemini AI provides actionable feedback that improves a user's chances of passing ATS filters

CHAPTER 8 :

CONCLUSION

The development and implementation of **CareerPulse** represent a successful synthesis of industrial engineering logic and modern generative artificial intelligence. This project was initiated to address a growing disparity in the digital age: while sectors like manufacturing have reached unprecedented levels of efficiency through automation, the individual professional journey remains bogged down by manual, repetitive, and cognitively taxing administrative hurdles. By drawing on the organizational and production efficiencies observed during my internship at **Solar Industries India Limited**, I have successfully demonstrated that the "production line" philosophy can be effectively applied to the career search lifecycle.

CareerPulse is not merely a web application; it is a distributed, intelligent ecosystem. Through the use of a high-performance FastAPI backend and a responsive Next.js frontend, the platform provides a seamless interface for complex backend operations. The integration of the Google Gemini-3-Flash model has proved that Large Language Models can do more than generate text—they can act as expert analysts, providing users with a 92% accuracy rate in resume parsing and ATS optimization. Furthermore, the Auto-Apply Agent, built on the Playwright framework, has fundamentally altered the quantitative dynamics of the job hunt. By reducing the average application time from 15 minutes to under 2 minutes—an efficiency gain of approximately 86.4 % the platform empowers job seekers to compete in a high-volume market with industrial precision.

Ultimately, this project proves that the burden of the modern job search can be significantly alleviated through the strategic application of autonomous agents. CareerPulse transforms what was once a chaotic and demoralizing manual struggle into a structured, data-driven, and automated pipeline. It stands as a testament to the potential of AI to democratize productivity, allowing individuals to spend less time on paperwork and more time on professional growth and skill acquisition.

CHAPTER 9:

FUTURE SCOPE

While the current iteration of CareerPulse provides a robust and functional foundation, the horizon for future development is expansive. The modular nature of the system architecture allows for several strategic enhancements that can move the platform from an "assistant" to a fully "autonomous career manager."

1. Expansion of the Automation Ecosystem The current Auto-Apply Agent is primarily optimized for LinkedIn's ecosystem. Future work will involve developing specialized adapters for a wider range of global job portals, including Indeed, Glassdoor, and Naukri. This will require the implementation of a universal "Field-Mapping Engine" that can intelligently identify form requirements across diverse UI structures, regardless of the underlying HTML nomenclature.

2. Hyper-Personalization through Generative AI The next phase of development will focus on "Deep Personalization." By utilizing the reasoning capabilities of the Gemini 1.5 Pro or future models, CareerPulse will be able to generate bespoke cover letters and tailor resume summaries for every specific job description it encounters. This ensures that the speed of automation does not come at the cost of the "personal touch" that recruiters value.

3. Behavioral Stealth and Security As job portals implement increasingly sophisticated bot-detection systems, future scope includes the development of Human-Mimicry Algorithms. This involves simulating non-linear mouse movements, variable typing speeds, and randomized interval delays. Research into biometric simulation will be critical to ensuring that the Auto-Apply Agent remains indistinguishable from a human user, maintaining long-term platform viability.

4. Predictive Career Pathing By integrating historical hiring data and market trends, future versions of the platform could provide Predictive Analytics. Instead of just applying to jobs, the AI could suggest specific skills the user should learn to increase their salary potential by a specific percentage, essentially acting as a long-term strategic career consultant.

5. Conversational Interview Preparation Finally, we aim to integrate a Voice-AI Mock Interview module. This feature would allow users to practice their interviewing skills with an AI that knows their resume and the job description perfectly, providing real-time feedback on tone, technical accuracy, and confidence.

CHAPTER 10 :

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Photo with Institute Guide

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Member: Aniket Rahile

Project Title: CareerPulse : Intelligent Career Recommendation Platform Using Resume Analytics

Photo with Industry Guide

Institute Guide: Ms.Ramadevi Salunkhe

Member: Aniket Rahile

Project Title: CareerPulse : Intelligent Career Recommendation Platform Using Resume Analytics